

Foundations of Protein structure prediction

[Protein structure prediction]

1.0 Content Block

Modeling of side-chain conformations Accurate packing of the amino acid side chains represents a separate problem in protein structure prediction. Methods that specifically address the problem of predicting side-chain geometry include dead-end elimination and the self-consistent mean field methods. The side chain conformations with low energy are usually determined on the rigid polypeptide backbone and using a set of discrete side chain conformations known as "rotamers". The methods attempt to identify the set of rotamers that minimize the model's overall energy. These methods use rotamer libraries, which are collections of favorable conformations for each residue type in proteins. Rotamer libraries may contain information about the conformation, its frequency, and the standard deviations about mean dihedral angles, which can be used in sampling. Rotamer libraries are derived from structural bioinformatics or other statistical analysis of side-chain conformations in known experimental structures of proteins, such as by clustering the observed conformations for tetrahedral carbons near the staggered (60° , 180° , -60°) values. Rotamer libraries can be backbone-independent, secondary-structure-dependent, or backbone-dependent. Backbone-independent rotamer libraries make no reference to backbone conformation, and are calculated from all available side chains of a certain type (for instance, the first example of a rotamer library, done by Ponder and Richards at Yale in 1987). Secondary-structure-dependent libraries present different dihedral angles and/or rotamer frequencies for α $\{\displaystyle \backslash\alpha\}$ -helix, β $\{\displaystyle \backslash\beta\}$ -sheet, or coil secondary structures. Backbone-dependent rotamer libraries present conformations and/or frequencies dependent on the local backbone conformation as defined by the backbone dihedral angles ϕ $\{\displaystyle \backslash\phi\}$ and ψ $\{\displaystyle \backslash\psi\}$, regardless of secondary structure. The modern versions of these libraries as used in most software are presented as multidimensional distributions of probability or frequency, where the peaks correspond to the dihedral-angle conformations considered as individual rotamers in the lists. Some versions are based on very carefully curated data and are used primarily for structure validation, while others emphasize relative frequencies in much larger data sets and are the form used primarily for structure prediction, such as the Dunbrack rotamer libraries. Side-chain packing methods are most useful for analyzing the protein's hydrophobic core, where side chains are more closely packed; they have more difficulty addressing the looser constraints and higher flexibility of surface residues, which often occupy multiple rotamer conformations rather than just one.

2.0 Content Block

The peptide bonds in the chain are polar, i.e. they have separated positive and negative charges (partial charges) in the carbonyl group, which can act as hydrogen bond acceptor and in the NH group, which can act as hydrogen bond donor. These groups can therefore interact in the protein structure. Proteins consist mostly of 20 different types of L- α -amino acids (the proteinogenic amino acids). These can be classified according to the chemistry of the side chain, which also plays an important structural role. Glycine takes on a special position, as it has the smallest side chain, only one hydrogen atom, and therefore can increase the local flexibility in the protein structure. Cysteine in contrast can react with another cysteine residue to form one cystine and thereby form a cross link stabilizing the whole structure. Protein structure arises from a sequence of secondary structure elements, such as α helices and β sheets. In secondary structures, regular patterns of H-bonds are formed between the main chain NH and CO groups of spatially neighboring amino acids, and the amino acids have similar Φ and ψ angles.

3.0 Content Block

In the case of complexes of two or more proteins, where the structures of the proteins are known or can be predicted with high accuracy, protein–protein docking methods can be used to predict the structure of the complex. Information of the effect of mutations at specific sites on the affinity of the complex helps to understand the complex structure and to guide docking methods.

4.0 Content Block

Evolutionary covariation to predict 3D contacts As sequencing became more commonplace in the 1990s several groups used protein sequence alignments to predict correlated mutations and it was hoped that these coevolved residues could be used to predict tertiary structure (using the analogy to distance constraints from experimental procedures such as NMR). The assumption is when single residue mutations are slightly deleterious, compensatory mutations may occur to restabilize residue-residue interactions. This early work used what are known as local methods to calculate correlated mutations from protein sequences, but suffered from indirect false correlations which result from treating each pair of residues as independent of all other pairs. In 2011, a different, and this time global statistical approach, demonstrated that predicted coevolved residues were sufficient to predict the 3D fold of a protein, providing there are enough sequences available (>1,000 homologous sequences are needed). The method, EVfold, uses no homology modeling, threading or 3D structure fragments and can be run on a standard personal computer even for proteins with hundreds of residues. The accuracy of the contacts predicted using this and related approaches has now been demonstrated on many known structures and contact maps, including the prediction of experimentally unsolved transmembrane proteins.

5.0 Content Block

Protein structure and terminology Proteins are chains of amino acids joined together by peptide bonds. Many conformations of this chain are possible due to the rotation of the main chain about the two torsion angles ϕ and ψ at the $C\alpha$ atom. This conformational flexibility is responsible for differences in the three-dimensional structure of proteins.

6.0 Content Block

AI methods AlphaFold was one of the first AIs to predict protein structures. It was introduced by Google's DeepMind in the 13th CASP competition, which was held in 2018. AlphaFold relies on a neural network approach, which directly predicts the 3D coordinates of all non-hydrogen atoms for a given protein using the amino acid sequence and aligned homologous sequences. The AlphaFold network consists of a trunk which processes the inputs through repeated layers, and a structure module which introduces an explicit 3D structure. Earlier neural networks for protein structure prediction used LSTM.

7.0 Content Block

CASP, which stands for Critical Assessment of Techniques for Protein Structure Prediction, is a community-wide experiment for protein structure prediction taking place every two years since 1994. CASP provides with an opportunity to assess the quality of available human, non-automated methodology (human category) and automatic servers for protein structure prediction (server category, introduced in the CASP7). The CAMEO3D Continuous Automated Model EvaluatiOn Server evaluates automated protein structure prediction servers on a weekly basis using blind predictions for newly release protein structures. CAMEO publishes the results on its website.